BEARLYTICS

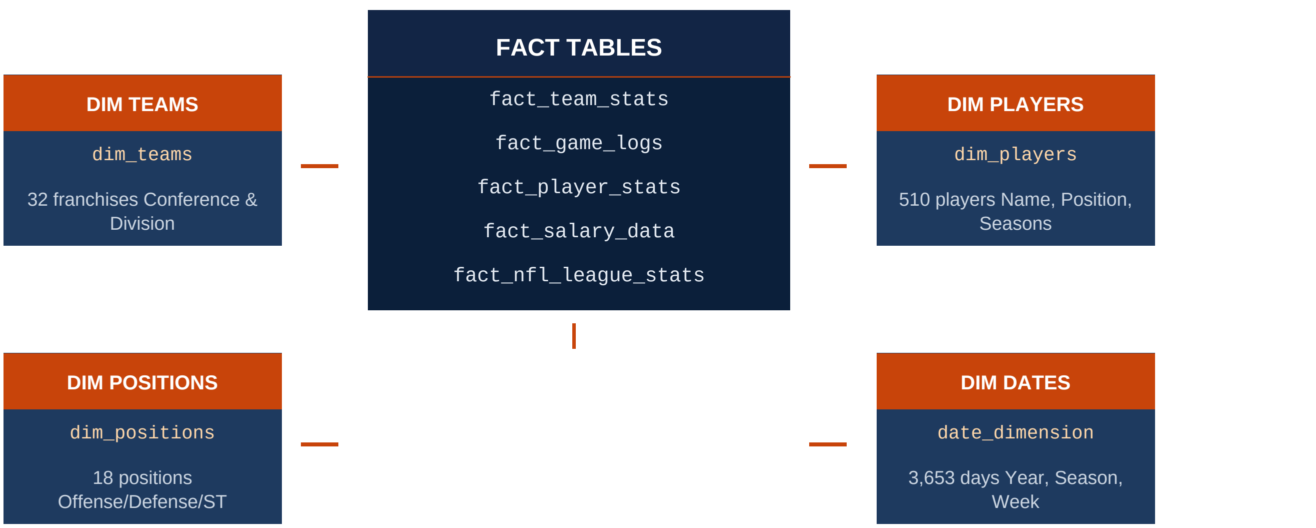
CHICAGO BEARS
ANALYTICS PLATFORM

PHASE 4 Power BI Development & DAX Modeling 2016 – 2025 NFL Seasons

Following the completion of the relational SQL database in Phase 3, Phase 4 focused on transforming validated analytical data into an interactive business intelligence solution using Microsoft Power BI. Rather than displaying raw tables or static charts, semantic modeling, DAX calculations, and interactive visualizations were developed to create a professional analytics application capable of answering executive-level business questions. The objective was to replicate the reporting layer used by business intelligence teams, where validated data is transformed into actionable insights through dynamic dashboards.

Step 1 — Power BI Data Model

The first step was importing the validated analytical model into Power BI and configuring relationships between fact and dimension tables. A star schema was implemented with one-to-many relationships from dimensions to fact tables, providing efficient filtering, simplified report development, and scalable analytical performance.



Step 2 — DAX Measure Development

With the data model complete, over 30 reusable DAX measures were created to calculate business metrics dynamically rather than relying on static columns. Measures update automatically in response to every filter, slicer, and cross-filter interaction.

Executive Performance

- fx Total Wins
- fx Total Losses
- fx Win Percentage
- fx Best Season
- fx Worst Season
- fx Average Wins

Offensive Production

- fx Points Scored
- fx Points Allowed
- fx Point Differential
- fx Passing Yards
- fx Rushing Yards
- fx Total Offensive Yards
- fx Turnover Differential

Roster Investment

- fx Total Contract Value
- fx Average Contract Value
- fx Largest Contract
- fx Guaranteed Money
- fx Cost Per Win
- fx Average Contract Length

Player Performance

- fx Passing Leaders
- fx Rushing Leaders
- fx Receiving Leaders
- fx Total Offensive Production

League Comparison

- fx League Average Wins
- fx Offensive Rank
- fx Defensive Rank
- fx Bears vs League Difference

Step 3 — Interactive Report Design

Rather than producing a single dashboard, Bearlytics was designed as a multi-page analytical application with each page focused on a distinct business question. Five report pages were developed.

01 **Executive Summary** — *"How has the franchise performed over the last decade?"*

KPI cards, win/loss trends, points scored and allowed, season-by-season performance summary

02 **Team Performance** — *"Which areas of team performance improved or declined?"*

Offensive and defensive production trends, year-over-year comparisons, point differential analysis

03 **Player Performance** — *"Which players drove offensive success?"*

Season-level passing, rushing, and receiving leaders; career production totals; positional contribution

04 **Roster Investment** — *"How effectively was roster spending converted into team performance?"*

Contract value distribution, positional investment breakdown, salary efficiency metrics

05 **Franchise Insights** — *"What are the key long-term insights from the Bears' performance?"*

Combined KPIs, league comparisons, ten-year trend synthesis, executive observations

Step 4 — Report Interactivity

Interactive reporting features were implemented throughout the dashboard to improve usability and support exploratory analysis. Users can move seamlessly between executive summaries and detailed performance analysis without modifying the underlying data.

Season Slicers	Filter all visuals by individual season or multi-season range
Position Filters	Isolate offensive, defensive, or special teams players
Cross-Filtering	Selecting any visual filters all related visuals on the page
Dynamic KPI Cards	Key metrics update in real time with every filter change
Drill-Through	Navigate from summary pages to detailed player or game records
Interactive Tooltips	Hover over any data point for contextual detail
Dynamic Titles	Chart headings update automatically to reflect active filters
Responsive Filtering	Filter state persists consistently across all five report pages

Step 5 — Dashboard Design

Consistent design standards were applied across every report page to improve readability and create a professional user experience. Every page follows a common design language aligned with Chicago Bears branding.

Bears Branding	Chicago Bears navy, orange, and white color palette
Typography	Consistent font sizing and weight hierarchy across all pages
KPI Cards	Executive metrics presented in standardized card format
Spacing & Alignment	Uniform padding, margins, and visual alignment throughout
Navigation Tabs	Interactive page navigation bar on every report page
Visual Hierarchy	Primary insights lead, supporting detail follows
Minimalist Layout	White space used deliberately to reduce visual clutter

Phase 4 Deliverables

At the completion of Phase 4, Bearlytics contains a fully interactive Power BI analytics solution built on a validated SQL data model.

Deliverable	Description	Type
Power BI Report (.pbix)	Complete interactive analytics application	Power BI
5 Dashboard Pages	Executive, Team, Player, Roster, Franchise	Reports

30+ DAX Measures	Dynamic business calculations	DAX
Star Schema Model	9-table relational model with relationships	Model
Interactive Filters	Season and position slicers across all pages	UX
KPI Cards	Executive performance metrics, dynamic	Visual
Business Visualizations	30+ interactive charts and tables	Visual

5	30+	9	30+	Visualiza

Outcome

Phase 4 transforms Bearlytics from a relational analytical database into a fully interactive business intelligence application. By developing a semantic data model, creating reusable DAX measures, and designing executive-focused dashboards, the project demonstrates the complete reporting layer of a modern analytics workflow.

Rather than presenting isolated charts, the dashboard provides decision-makers with a unified platform for exploring franchise performance, player production, roster investment, and league comparisons across ten NFL seasons. The result mirrors the type of self-service reporting solution commonly delivered by business intelligence analysts using Microsoft Power BI.

Project Roadmap

Phase 1	**Phase 2**	**Phase 3**	**Phase 4**	**Phase 5**	**Phase 6**
Data Collection & Engineering	Data Validation & Modeling	SQL Database & Business Analysis	Power BI & DAX Modeling	Executive Dashboard	GitHub Portfolio & Presentation

Green = complete. Orange = current phase. Remaining phases cover executive dashboard development and GitHub portfolio presentation.